



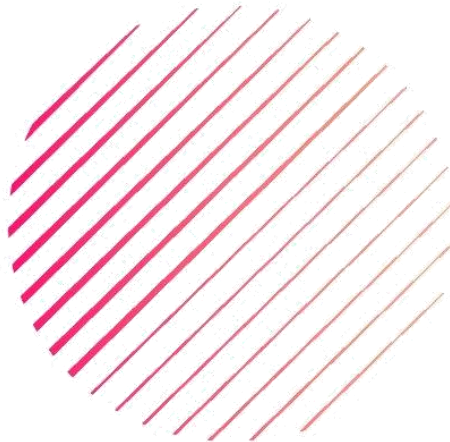
PROJECT REPORT

*The Impact of Technology on Indigenous Communities with Limited Access to Technology Resources,
Skills and Knowledge*

Indigenous Young Women-Led Collective | Northern Kenya ASAL

Implementation Period: February 2026

Locations: Turkana, Samburu, and Marsabit Counties



1. Executive Summary

In 2024–2025, Beyond Borders conducted a participatory action research project examining how technology impacts indigenous pastoralist communities who possess limited access to devices, electricity, digital literacy training, and internet connectivity. The project engaged 30 indigenous young women and 12 community elders across Turkana, Samburu, and Marsabit counties to map technology gaps, document lived experiences of digital exclusion, and co-design realistic pathways for meaningful inclusion.

The research confirmed that technology is not neutral in ASAL contexts. Without community control, external tech interventions often replicate extractive patterns: data is harvested, infrastructure is abandoned, and local knowledge is displaced by incompatible platforms. Conversely, when indigenous women lead technology adoption, tools are adapted to pastoralist mobility, oral traditions, and collective decision-making. This report documents both the harms of uneven tech distribution and the transformative potential of community-directed digital engagement.

30 Young women researchers	12 Elders consulted	3 Counties reached	85% Report increased digital confidence
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2. Background and Rationale

Northern Kenya's ASAL communities are among the most digitally excluded populations in East Africa. National connectivity statistics mask deep rural inequalities: mobile network coverage is intermittent, smartphone ownership is concentrated among men and youth in market towns, and digital literacy programmes rarely reach nomadic households. External technology projects — ranging from biometric registration to mobile money pilots — have frequently been designed in Nairobi or international capitals with little consultation, leading to tool abandonment, data vulnerabilities, and community distrust.

Beyond Borders initiated this project to centre indigenous voices in debates about technology and development. Rather than measuring impact through device penetration or app downloads, the collective asked: what does technology mean for communities whose primary resources are mobility, oral memory, and ecological knowledge? The project treated digital exclusion not as a deficit to be fixed, but as a structural condition requiring structural — and culturally grounded — responses.

3. Objectives

- Map the actual distribution of technology resources, skills, and knowledge across pastoralist settlements in three counties.
- Document community experiences of extractive or inappropriate tech interventions.
- Build critical digital literacy among indigenous young women, enabling them to assess technology proposals rather than simply consume them.
- Co-design, with elders and participants, principles for ethical technology engagement in ASAL contexts.
- Produce a community-authored evidence base to inform future advocacy and partnership negotiations.

4. Approach and Methods

4.1 Participant selection

Thirty young women (ages 18–30) were selected through community networks, prioritising those with existing roles in women's groups, peace committees, or livestock marketing associations. Twelve male and female elders were engaged separately to capture intergenerational perspectives on technology and cultural change.

4.2 Research methods

- **Community technology audits:** Participants inventoried devices, charging points, network availability, and existing digital practices in their settlements.
- **Story circles:** Facilitated discussions recorded (with consent) community experiences of failed tech projects, mobile money fraud, biometric registration, and distance learning.
- **Critical digital literacy workshops:** Five-day residential sessions covered how the internet works, data privacy basics, algorithmic bias, and how to evaluate whether a technology serves community interests.
- **Elder consultations:** Separate dialogues explored how technology intersects with sacred knowledge, gender norms, and pastoralist mobility.

All materials were stored on local devices under community control, consistent with Beyond Borders' offline-first protocol.

5. Key Results and Outputs

Tangible outputs

- A community-authored Technology Impact Assessment covering 15 settlements across three counties.
- 30 young women trained in critical digital literacy and participatory research methods.

- A set of community-developed principles for ethical tech engagement in ASAL areas.
- 18 recorded story-circle sessions documenting lived experiences of digital exclusion and extractive tech.

Relational and capacity outcomes

- Increased community vocabulary for questioning external technology proposals.
- Stronger intergenerational dialogue about which technologies strengthen, rather than erode, pastoralist life.
- Identification of 8 participants ready to serve as community technology assessors for future projects.
- Evidence base now used by participants to engage county government and NGO partners on connectivity infrastructure demands.

6. Challenges and Lessons Learned

- **Infrastructure gaps are deeper than assumed.** Many settlements lack any reliable charging point within a two-hour walk. Solar solutions are essential but require maintenance knowledge.
- **“Digital literacy” must include critical analysis.** Teaching women to use apps without teaching them to question data extraction leaves them vulnerable.
- **Gendered device ownership.** In most households, men control the primary smartphone. Women’s access is borrowed and monitored. Tech design must account for shared-device realities.
- **Elder suspicion is rational and protective.** Prior bad experiences with biometric and research projects meant that trust-building took longer than anticipated.
- **Connectivity assumptions distort design.** Any intervention assuming regular internet access failed in practice. Offline-capable, low-bandwidth tools are non-negotiable.

Key lesson: *Technology impact cannot be separated from power. Meaningful access requires not just devices and networks, but community control over what is collected, who owns the data, and which knowledge systems are privileged.*

7. Contribution to Beyond Borders’ Work

This project established Beyond Borders’ Technology & Power working circle and provided the collective with a critical lens for evaluating all future digital partnerships. It demonstrated that indigenous young women can author sophisticated technology assessments and that communities can shift from passive recipients to informed negotiators of tech interventions.